

APPENDIX B

EXCEPTION PROCESSING REFERENCE

This appendix provides a quick reference for system programmers who are already familiar with the stack frames. For more detail, please refer to the appropriate userOs manual.

B.1 EXCEPTION VECTOR ASSIGNMENTS FOR THE M68000 FAMILY

Table B-1 lists all vector assignments up to and including the MC68040 and its derivatives. Many of these vector assignments are processor specific. For instance, vector 13, the coprocessor protocol violation vector, only applies to the MC68020, MC68EC020, MC68030, and MC68EC030. Refer to the appropriate user's manual to determine which exception type is applicable to a specific processor.

Table B-1. Exception Vector Assignments for the M68000 Family

Vector Number(s)	Vector Offset (Hex)	Assignment
0	000	Reset Initial Interrupt Stack Pointer
1	004	Reset Initial Program Counter
2	008	Access Fault
3	00C	Address Error
4	010	Illegal Instruction
5	014	Integer Divide by Zero
6	018	CHK, CHK2 Instruction
7	01C	FTRAPcc, TRAPcc, TRAPV Instructions
8	020	Privilege Violation
9	024	Trace
10	028	Line 1010 Emulator (Unimplemented A- Line Opcode)
11	02C	Line 1111 Emulator (Unimplemented F-Line Opcode)
12	030	(Unassigned, Reserved)
13	034	Coprocessor Protocol Violation
14	038	Format Error
15	03C	Uninitialized Interrupt
16–23	040–05C	(Unassigned, Reserved)
24	060	Spurious Interrupt
25	064	Level 1 Interrupt Autovector
26	068	Level 2 Interrupt Autovector
27	06C	Level 3 Interrupt Autovector
28	070	Level 4 Interrupt Autovector
29	074	Level 5 Interrupt Autovector
30	078	Level 6 Interrupt Autovector
31	07C	Level 7 Interrupt Autovector
32–47	080–0BC	TRAP #0 D 15 Instruction Vectors
48	0C0	FP Branch or Set on Unordered Condition
49	0C4	FP Inexact Result
50	0C8	FP Divide by Zero
51	0CC	FP Underflow
52	0D0	FP Operand Error
53	0D4	FP Overflow
54	0D8	FP Signaling NAN
55	0DC	FP Unimplemented Data Type (Defined for MC68040)
56	0E0	MMU Configuration Error
57	0E4	MMU Illegal Operation Error
58	0E8	MMU Access Level Violation Error
59–63	0ECD0FC	(Unassigned, Reserved)
64–255	100D3FC	User Defined Vectors (192)

B.2 EXCEPTION STACK FRAMES

Figures B-1 through B-15 illustrate all exception stack frames for the M68000 family..

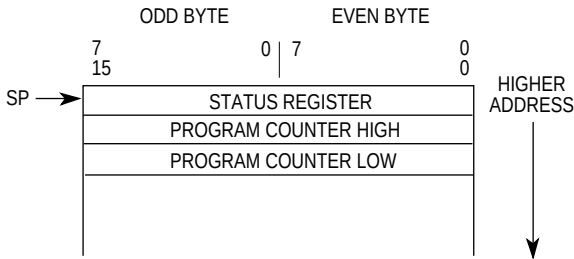


Figure B-1. MC68000 Group 1 and 2 Exception Stack Frame

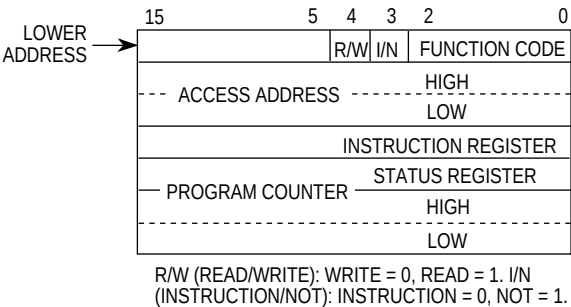


Figure B-2. MC68000 Bus or Address Error Exception Stack Frame

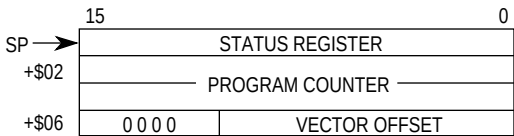


Figure B-3. Four-Word Stack Frame, Format \$0

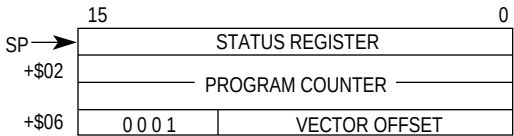


Figure B-4. Throwaway Four-Word Stack Frame, Format \$1

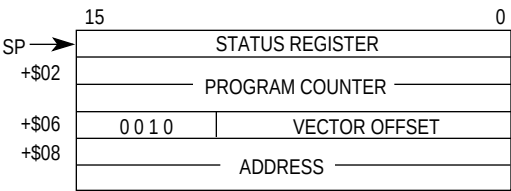


Figure B-5. Six-Word Stack Frame, Format \$2

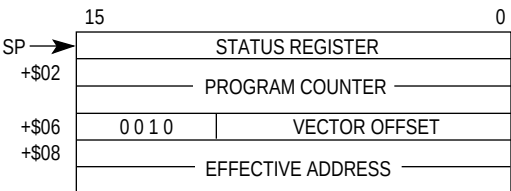


Figure B-6. MC68040 Floating-Point Post-Instruction Stack Frame, Format \$3

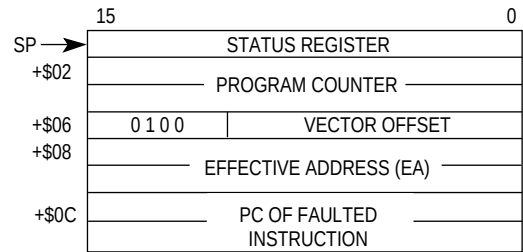


Figure B-7. MC68EC040 and MC68LC040 Floating-Point Unimplemented Stack Frame, Format \$4

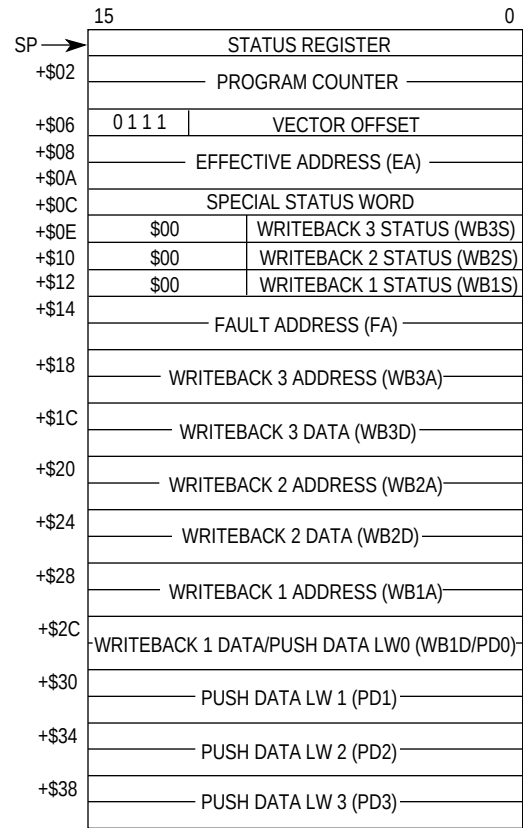
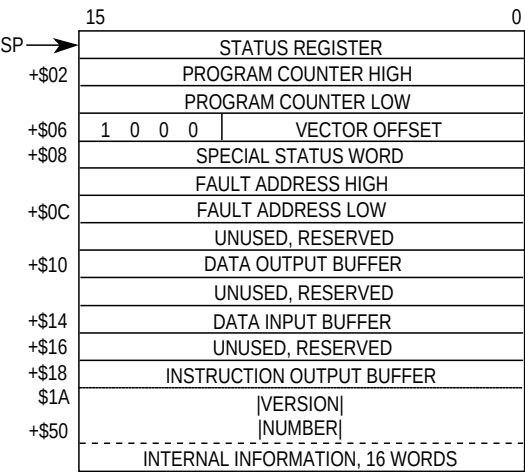


Figure B-8. MC68040 Access Error Stack Frame, Format \$7



NOTE: The stack pointer decrements by 29 words, although only 26 words of information actually write to memory. Motorola reserves the three additional words for future use.

Figure B-9. MC68010 Bus and Address Error Stack Frame, Format \$8

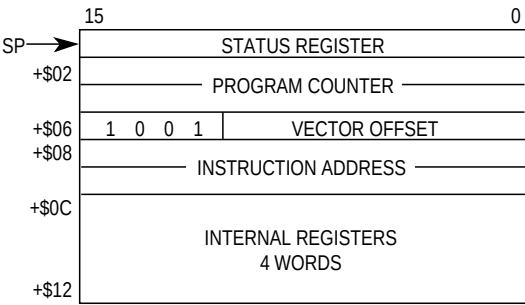


Figure B-10. MC68020 Bus and MC68030 Coprocessor Mid-Instruction Stack Frame, Format \$9

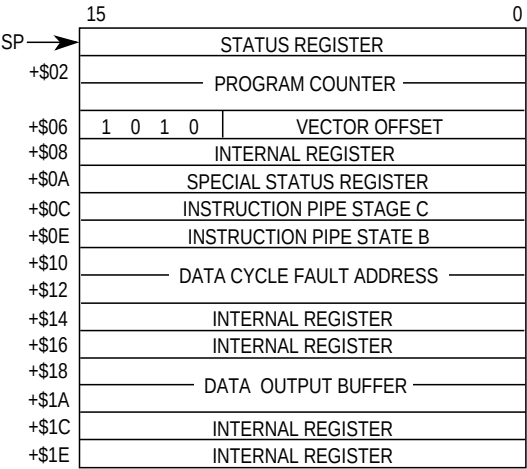


Figure B-11. MC68020 and MC68030 Short Bus Cycle Stack Frame, Format \$A

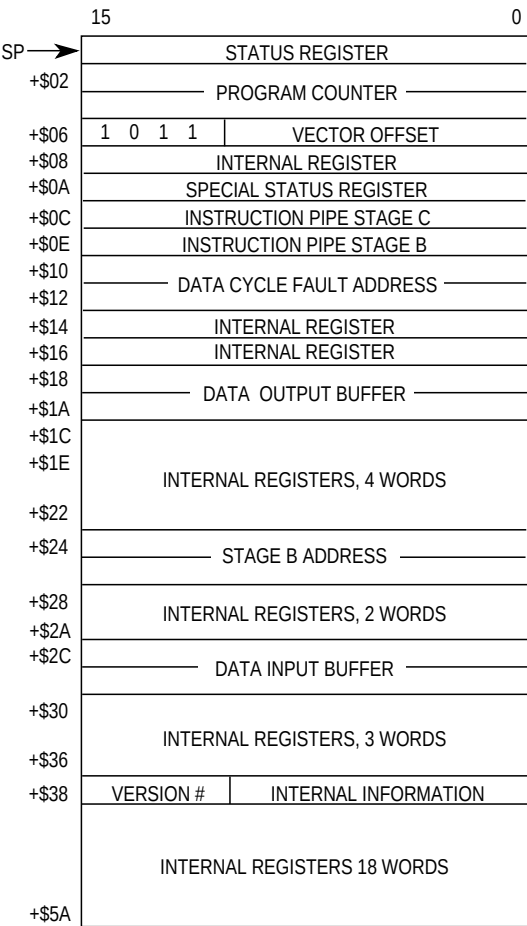


Figure B-12. MC68020 and MC68030 Long Bus Cycle Stack Frame, Format \$B

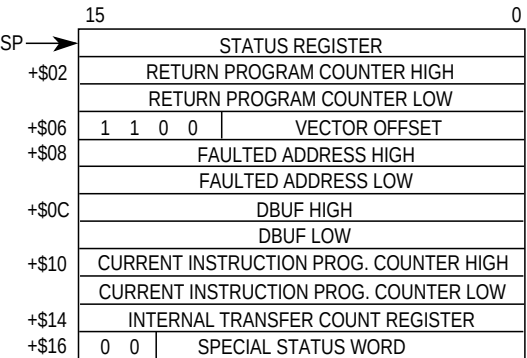


Figure B-13. CPU32 Bus Error for Prefetches and Operands Stack Frame, Format \$C

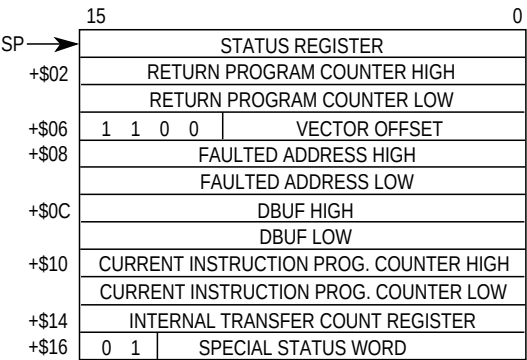


Figure B-14. CPU32 Bus Error on MOVEM Operand Stack Frame, Format \$C

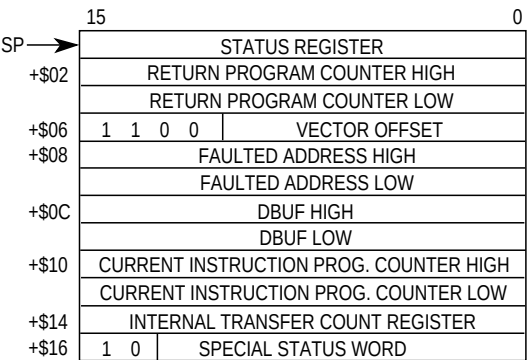


Figure B-15. CPU32 Four- and Six-Word Bus Error Stack Frame, Format \$C

B.3 FLOATING-POINT STACK FRAMES

Figures B-16 through B-23 illustrate floating-point stack frames for the MC68881/MC68882 and the MC68040.

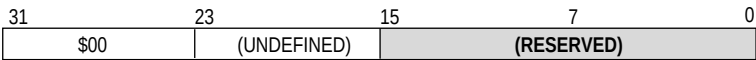


Figure B-16. MC68881/MC68882 and MC68040 Null Stack Frame

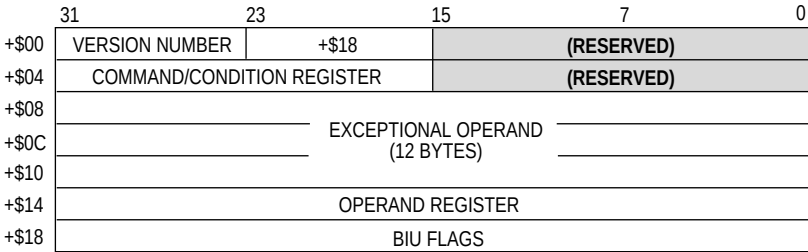


Figure B-17. MC68881 Idle Stack Frame

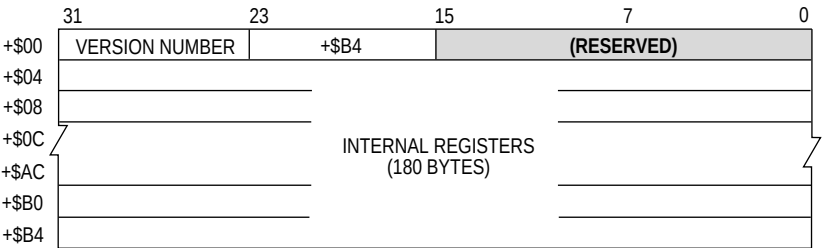


Figure B-18. MC68881 Busy Stack Frame

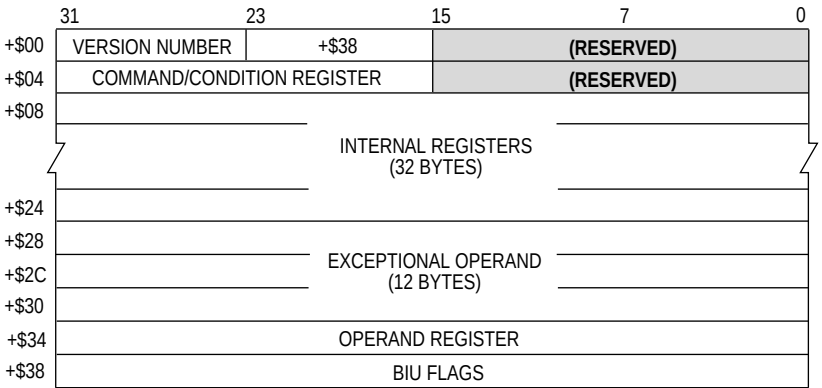


Figure B-19. MC68882 Idle Stack Frame

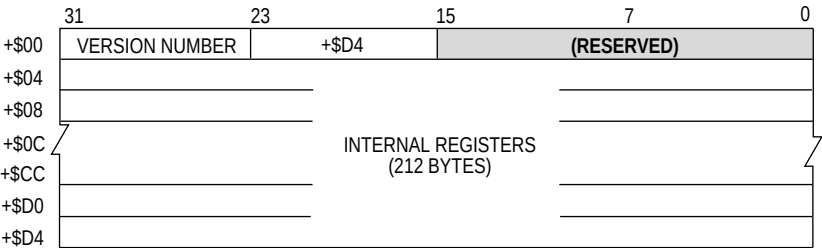


Figure B-20. MC68882 Busy Stack Frame

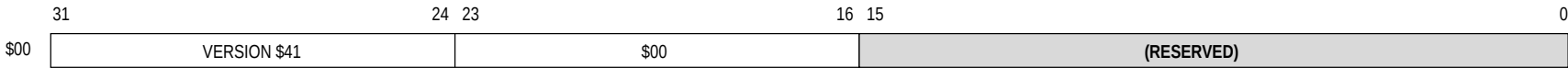
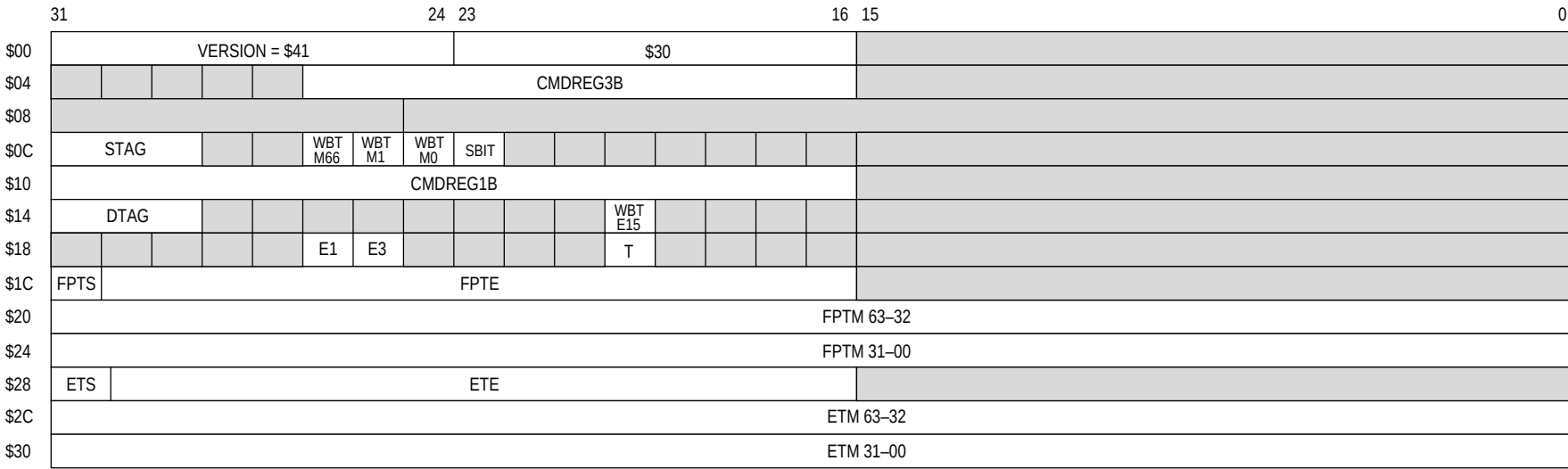
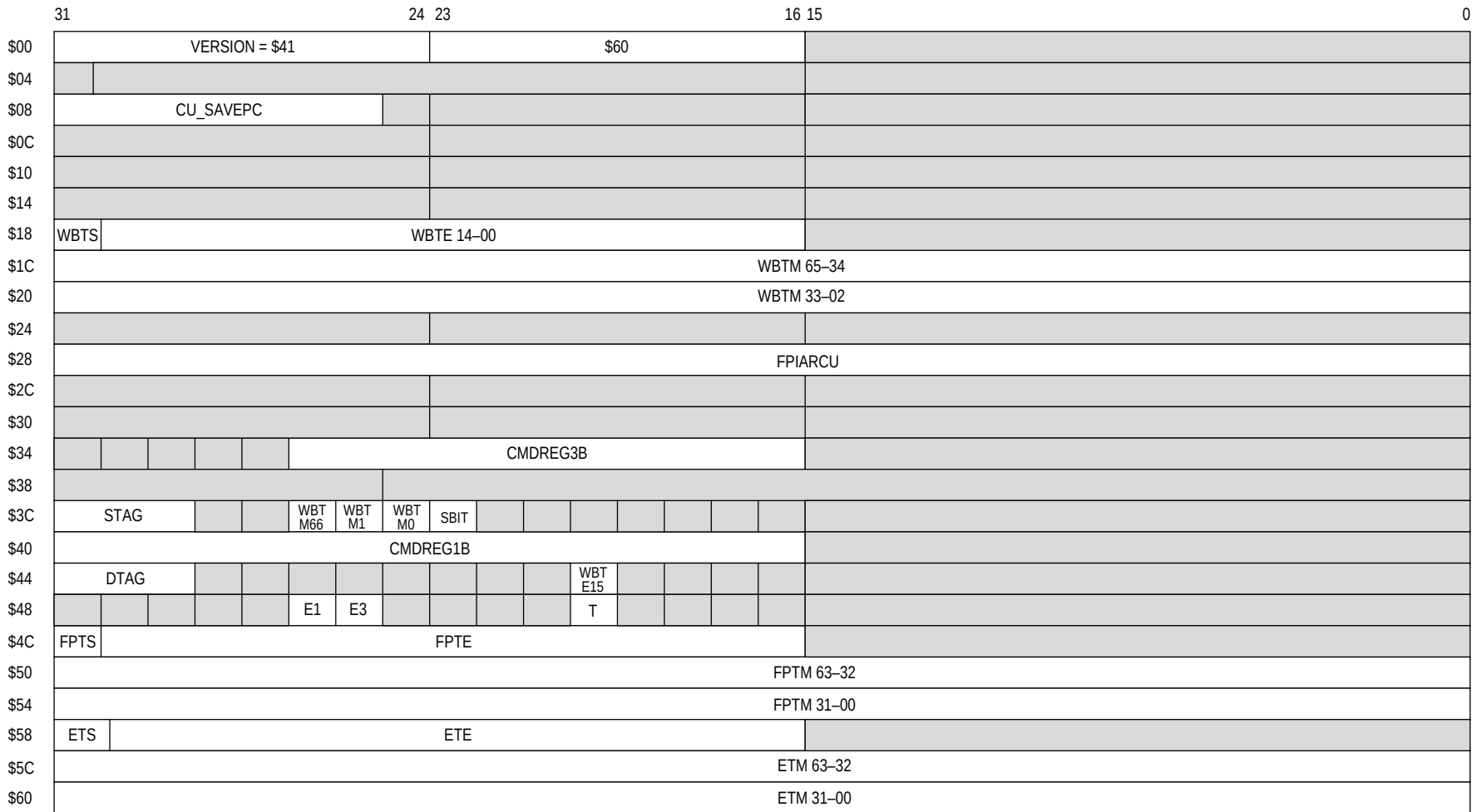


Figure B-21. MC68040 Idle Stack Frame



 Reserved

Figure B-22. MC68040 Unimplimented Instruction Stack Frame



 Reserved

Figure B-23. MC68040 Busy Stack Frame